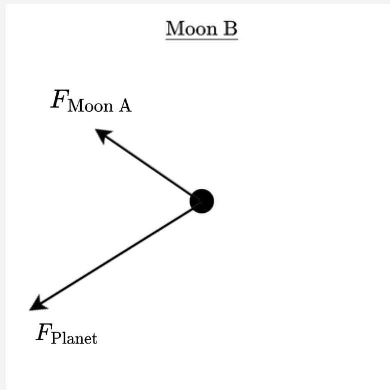
AP1 Practice Exam #1
FRQ Rubric

Question 1 — 10 Points

Part A

i.

Sample Response



$+1:$ For drawing an up-left arrow labeled as the force from moon A
$+1:$ For drawing a down-left arrow labeled as the force from the planet

$-1:$ For each extra arrow
$-1:$ For each arrow labeled as g or G
$-1:$ If the angle between $F_{\text{Moon A}}$ & F_{Planet} is not roughly 60°

ii.

$+1:$ $\sum F_c = \frac{mv^2}{r}$

$+1:$ $F_M + F_A\cos(60^\circ) = \frac{mv^2}{d}$; $F = G\frac{m_1 m_2}{r^2}$

$$\frac{GMm}{d^2} + \frac{Gm^2}{d^2}\cos(60^\circ) = \frac{mv^2}{d} \quad ; \quad \cos(60^\circ) = \frac{1}{2}$$

$$\frac{GMm}{d^2} + \frac{Gm^2}{2d^2} = \frac{mv^2}{d}$$

$$\frac{Gm}{d^2}\left(M + \frac{m}{2}\right) = \frac{mv^2}{d}$$

$$\frac{G}{d}\left(M + \frac{m}{2}\right) = v^2$$

$+1:$ $v = \sqrt{\frac{G}{d}\left(M + \frac{m}{2}\right)}$

iii.

$$v = \frac{\Delta x}{\Delta t} \qquad \begin{cases} \Delta x = 2\pi d \text{ (circumference)} \\ \Delta t = T \text{ (orbital period)} \end{cases}$$

$$v = \frac{2\pi d}{T}$$

$+1:$ $T = \frac{2\pi d}{v}$ $\qquad \begin{cases} \text{Carry over applicable from A(ii)} \\ v = \sqrt{\frac{G}{d}\left(M + \frac{m}{2}\right)} \end{cases}$

$$T = \frac{2\pi d}{\sqrt{\frac{G}{d}\left(M + \frac{m}{2}\right)}}$$

$$T = 2\pi d\sqrt{\frac{d}{G\left(M + \frac{m}{2}\right)}}$$

$+1:$ $T = 2\pi\sqrt{\frac{d^3}{G\left(M + \frac{m}{2}\right)}}$

Part B

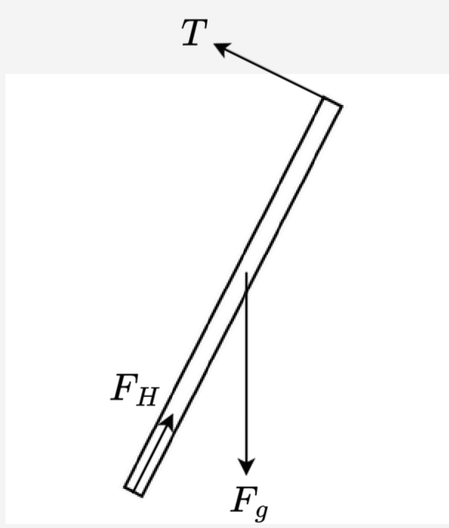
+1 : For stating $Ug > U_{g0}$
+1 : For indicating that the gravitational forces between the masses are attractive, meaning the potential energy is a negative value
+1 : For indicating that increasing the separation distance moves the system's potential energy closer to zero

−1 : For stating that the potential energy of the bound configuration is positive
−1 : For a justification that relies on a mathematical expression or derivation

Question 2 — 12 Points

Part A

Sample Response



+1 : For drawing a downward arrow labeled as the force from gravity, starting at the center of mass of the drawbridge
+1 : For drawing an up-left arrow labeled as the tension force from the cable, oriented perpendicular to the drawbridge at its outer end
+1 : For drawing an arrow at the axle pivot representing the force from the hinge

−1 : For each extra arrow or component drawn
−1 : For each arrow not starting on the point at which the force is exerted
−1 : If the force of gravity is labeled as g or G

Note: The relative lengths of the arrows should not be considered when scoring, as the magnitudes of F_H and F_T shift depending on the angle θ_0.

Part B

+1 : $\tau_{\text{net}} = 0$

+1 : $\tau_{\text{cable}} - \tau_{\text{gravity}} = 0$

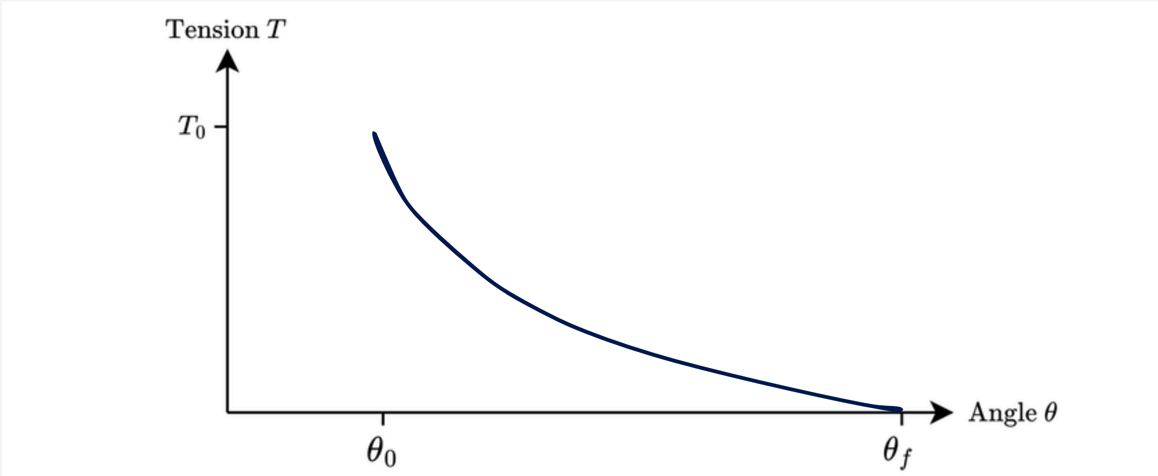
+1 : $T_0 L - Mg\left(\frac{L}{2}\right)\sin\theta_0 = 0$

$T_0 L = \frac{1}{2} MgL\sin\theta_0$

+1 : $T_0 = \frac{1}{2} Mg\sin\theta_0$

Part C

Sample Response



+1 : For a graph that starts at the point (θ_0, T_0)

+1 : For a curve that is concave down & decreasing throughout the entire interval from θ_0 to θ_f

+1 : For a curve that ends at the point $(\theta_0, 0)$

Part D

+1 : For stating that $T_{\text{new}} > T_{\text{uniform}}$

+1 : For justifying that concentrating the mass further from the axle increases the gravitational torque, requiring a greater tension torque to balance it

Question 3 — 10 Points

Part A

i.

+1 : For stating that both the distance traveled by the combined dart-cart system using the meterstick and the time interval taken to cover that distance using the stopwatch must be measured to determine the final velocity.

ii.

+1 : For stating to repeat the distance and time measurements across multiple identical trials at each launch speed setting to calculate a reliable average velocity

Part B

i.

+1 : For indicating that the vertical axis should plot the square of the final velocity and the horizontal axis should plot a constant value, or vice versa, to achieve a linear relationship that yields the final kinetic energy

ii.

+1 : For describing that the final kinetic energy is directly proportional to the slope of the line of best fit, related by the total combined mass of the system

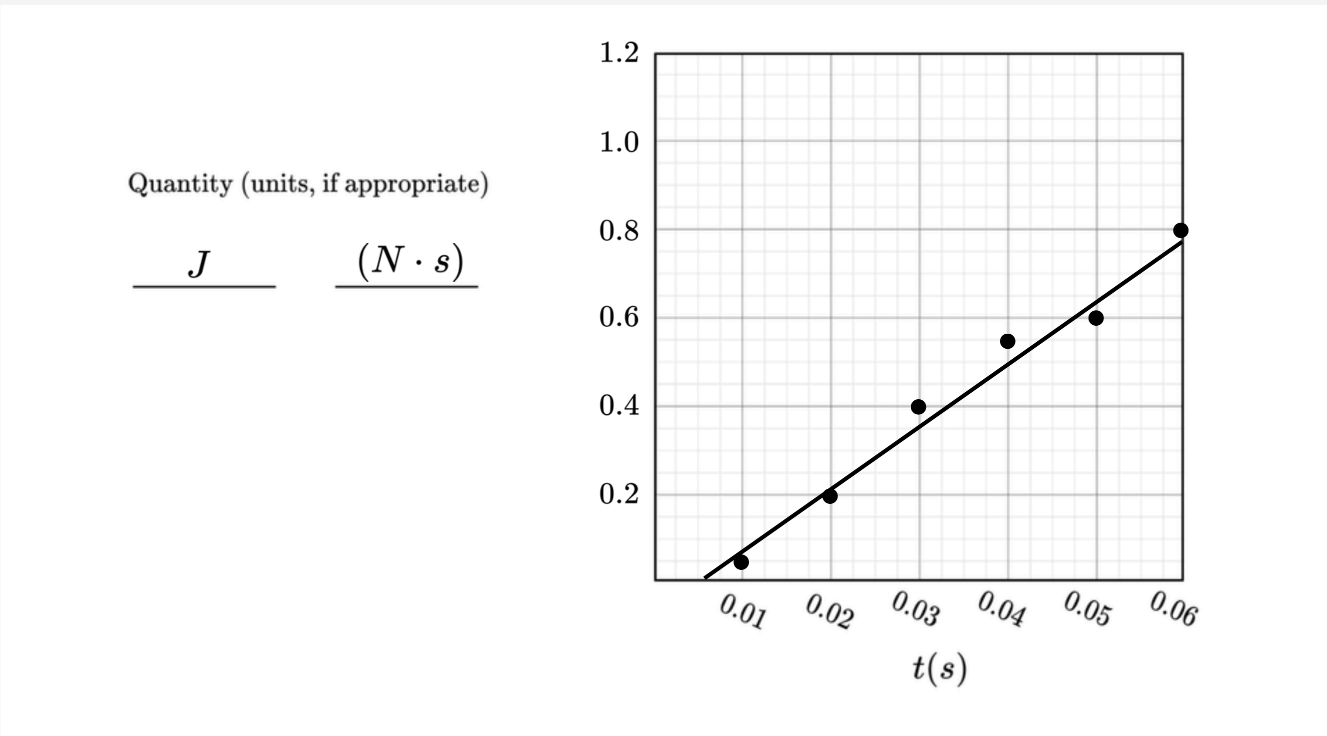
Part C

i.

+1 : For labeling the vertical axis as Impulse with the appropriate units of N · s to yield a linear relationship for analyzing average force over time

ii.

+1 : For correctly scaling and labeling the axes, as well as accurately plotting all data points from Table 1 on the provided grid



iii.

+1 : For drawing a single linear best-fit line that appropriately minimizes the distance to all plotted data points on the grid

Part D

$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$ $\begin{cases}\text{Calculate the experimental slope} \\ \text{using two points on the line}\end{cases}$

+1 : $\text{slope} = \frac{0.82 - 0.21}{0.06 - 0.02} = 15.25$

+1 : $J = \Delta p_{\text{wall}} = -\Delta p_{\text{dart}}$

$J = -(m_d[v(t)] - m_d v_0)$

$J = m_d(v_0 - [v(t)])$

$\frac{J}{m_d} = v_0 - [v(t)]$

$[v(t)] = v_0 - \frac{J}{m_d}$

$[v(t)] = v_0 - \frac{(\text{slope})t + \text{intercept}}{m_d}$

+1 : $[v(t)] = v_0 - \frac{15.25t - 0.10}{m_d}$

Question 4 — 8 Points

Part A

+1 : For indicating that $f < f_0$
+1 : For explaining that a longer pendulum increases the path length or arc length that the bob must travel during one full cycle of oscillation
+1 : For indicating that the component of gravitational acceleration along the path is unchanged, meaning a longer distance requires more time to traverse, thereby increasing the period and decreasing the frequency

Part B

+1 : For indicating Path B
+1 : For explaining that at the highest point of its swing, the bob has momentarily come to rest and has zero instantaneous velocity
+1 : For stating that once the string is cut, gravity is the only force acting on the bob, causing it to fall straight down under free fall acceleration

Part C

+1 : For explaining that due to conservation of energy, the bob is momentarily at rest at Point 2 and the elastic string returns to its unstretched length
+1 : For stating that the bob will fall straight down vertically because it has no initial velocity and gravity is the only acting force